Fine-tuning EvoDiff Model for Designing Thermostable Detergent Enzyme Sequences

2024/08/28

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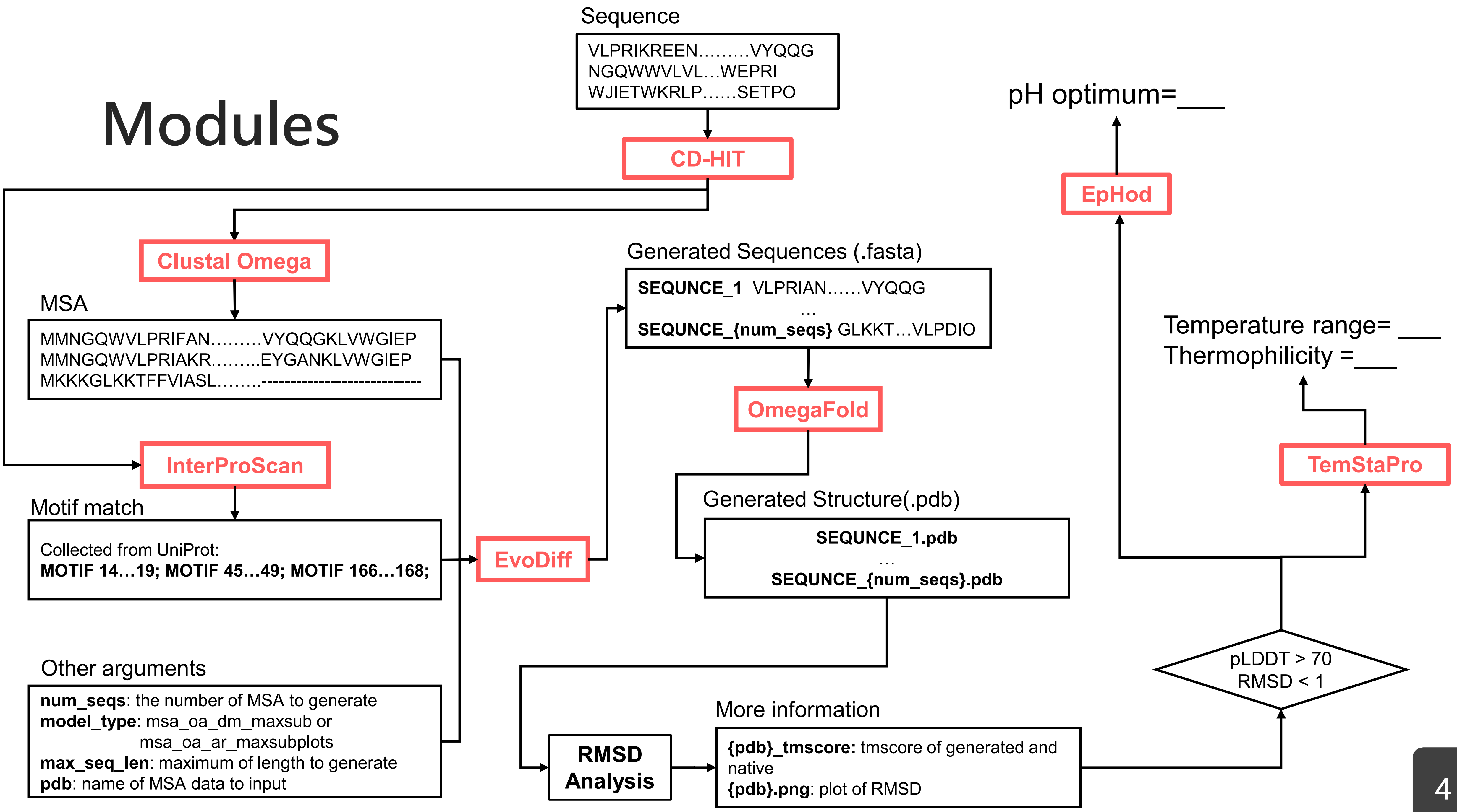
0 Introduction

Modules



Sequence
VLPRIKREEN.........VYQQG
NGQWWVLVL...WEPRI
WJIETWKRLP......SETPO
CD-HIT
Clustal Omega
MSA
MMNGQWVLPRIFAN.........VYQQGKLVWGIEP
MMNGQWVLPRIAKR.........EYGANKLVWGIEP
MKKKGLKKTFFVIASL........----------------------------
InterProScan
Motif match
Collected from UniProt:
MOTIF 14…19; MOTIF 45…49; MOTIF 166…168;
Other arguments
num_seqs: the number of MSA to generate
model_type: msa_oa_dm_maxsub or
msa_oa_ar_maxsubplots
max_seq_len: maximum of length to generate
pdb: name of MSA data to input
EvoDiff
Generated Sequences (.fasta)
SEQUNCE_1 VLPRIAN……VYQQG
…
SEQUNCE_{num_seqs} GLKKT…VLPDIO
OmegaFold
Generated Structure(.pdb)
SEQUNCE_1.pdb
…
SEQUNCE_{num_seqs}.pdb
RMSD Analysis
More information
{pdb}_tmscore: tmscore of generated and native
{pdb}.png: plot of RMSD
pLDDT > 70
RMSD < 1
EpHod
pH optimum=___
TemStaPro
Temperature range= ___
Thermophilicity =___

Models

Model	Details	Reference
Clustal Omega	A version, completely rewritten and revised in 2011, of the widely used Clustal series of programs for multiple sequence alignment.	*Sievers, Fabian and Desmond G. Higgins. “The Clustal Omega Multiple Alignment Package.” Methods in molecular biology 2231 (2021): 3-16 .*
CD-Hit	Widely used program for clustering and comparing protein or nucleotide sequences.	*Li, Weizhong and Adam Godzik. “Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences.” Bioinformatics 22 13 (2006): 1658-9 .* *Fu, Limin, Beifang Niu, Zhengwei Zhu, Sitao Wu and Weizhong Li. “CD-HIT: accelerated for clustering the next-generation sequencing data.” Bioinformatics 28 (2012): 3150 - 3152.*
EvoDiff	Able to generate diverse protein sequences in both conditional and unconditional states, predict their structures, and use OmegaFold to evaluate the quality.	*Alamdari, Sarah, Nitya Thakkar, Rianne van den Berg, Alex X. Lu, Nicolo Fusi, Ava P. Amini and Kevin Kaichuang Yang. “Protein generation with evolutionary diffusion: sequence is all you need.” bioRxiv (2023): n. pag.*
OmegaFold	Predict structure of protein sequence	*Wu, Rui Min, Fan Ding, Rui Wang, Rui Shen, Xiwen Zhang, Shitong Luo, Chenpeng Su, Zuofan Wu, Qi Xie, Bonnie Berger, Jianzhu Ma and Jian Peng. “High-resolution de novo structure prediction from primary sequence.” bioRxiv (2022): n. pag.*
InterProScan	• Scan motif for sequence. • Released on 25 July 2024: InterProScan 5.69-101.0	*Jones, Philip, David Binns, Hsin-Yu Chang, Matthew Fraser, Weizhong Li, Craig McAnulla, Hamish McWilliam, John Maslen, Alex L. Mitchell, Gift Nuka, Sebastien Pesseat, Antony F. Quinn, Amaia Sangrador-Vegas, Maxim Scheremetjew, Siew-Yit Yong, Rodrigo Lopez and Sarah Hunter. “InterProScan 5: genome-scale protein function classification.” Bioinformatics 30 (2014): 1236 - 1240.*
TemStaPro	Predict protein thermostability using embeddings generated by protein language models (pLMs) from an input protein sequence.	*Pudžiuvelytė, Ieva, Kliment Olechnovič, Egle Godliauskaite, Kristupas Sermokas, Tomas Urbaitis, Giedrius Gasiunas and Darius Kazlauskas. “TemStaPro: protein thermostability prediction using sequence representations from protein language models.” Bioinformatics 40 (2024): n. pag.*
EpHod	Learns structural and biophysical features from sequence data alone to predict enzyme pH_{opt} from protein sequence.	*Gado, Japheth E., Matthew Knotts, Ada Y. Shaw, Debora S. Marks, Nicholas Paul Gauthier, Christensen Carsten Sander and Gregg T. Beckham. “Deep learning prediction of enzyme optimum pH.” bioRxiv (2023): n. pag.*

1 Method

Dataset

Data name	Description
detergent-papers	Papers on detergent-compatible enzymes (amylase, cellulase, lipase, mannanase, protease, others). Includes EC numbers, organisms, optimal pH and temperature. Used to collect UniProt sequence data.
detergent-motif	Sequence data by EC number, with motif positions, optimal pH and temperature.
detergent-enzyme	Dataset for EvoDiff fine-tuning. 644 representative detergent-compatible sequences, split into training and test sets.

Dataset

- Available on Hugging Face Hub

Datasets 3

lun610200/detergent-enzyme
Viewer • Updated about 20 hours ago • 643

lun610200/detergent-motif
Viewer • Updated 1 day ago • 1.79k

lun610200/detergent-papers
Viewer • Updated 1 day ago • 153

Data Collection

Collection from Literature

To find identified microorganisms for extracting detergent enzymes and specific catalytic reaction EC numbers

Sequence Collection from UniProt

Use EC numbers and microorganisms found in papers as filters to Select protein sequence Data from UniProt

By EC and microorganism

It might be more accurate detergent enzyme data, but the quantity is limited

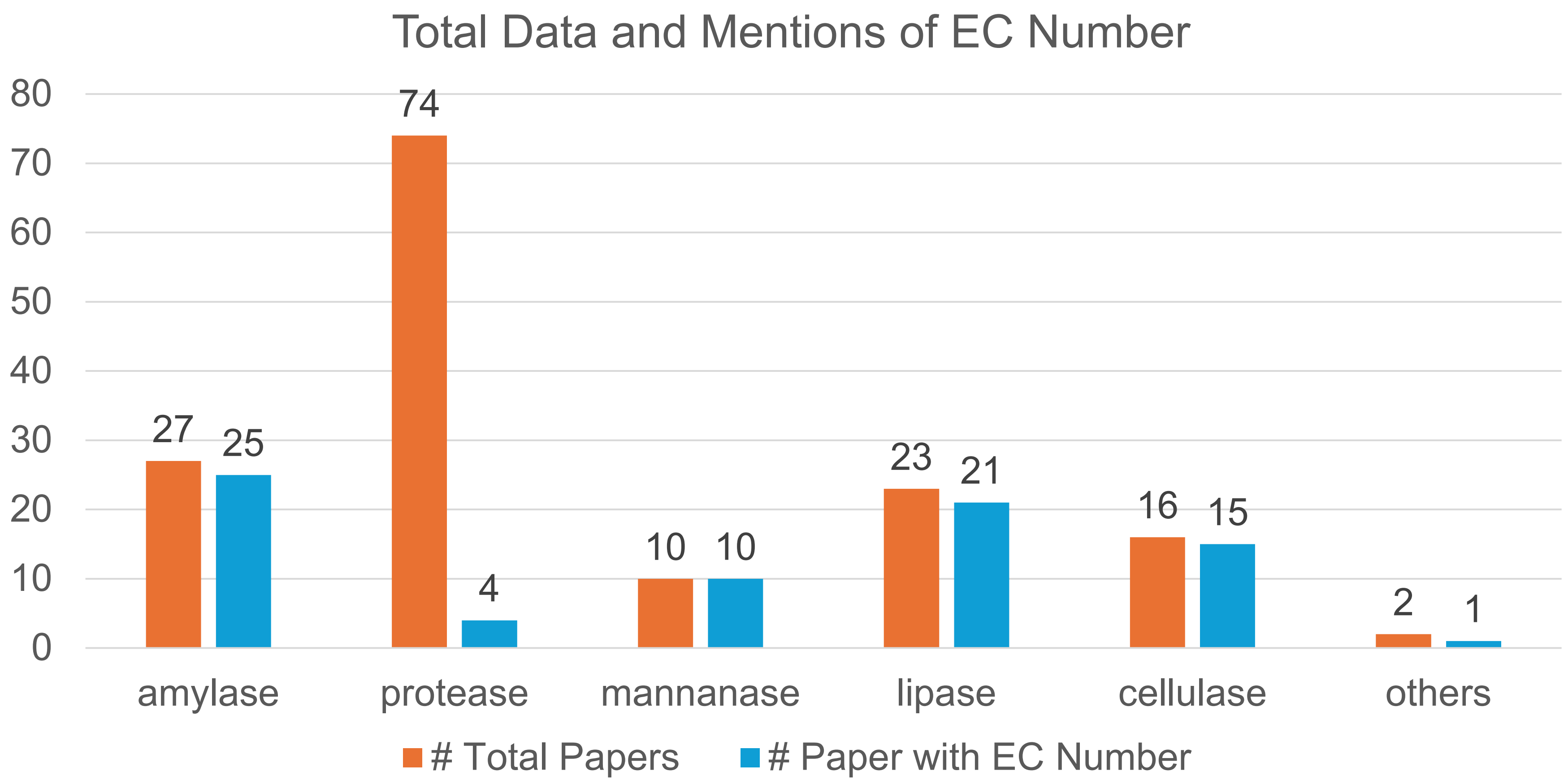
By EC and species

Expanding the scope from specific microorganisms to species can yield more protein sequence data

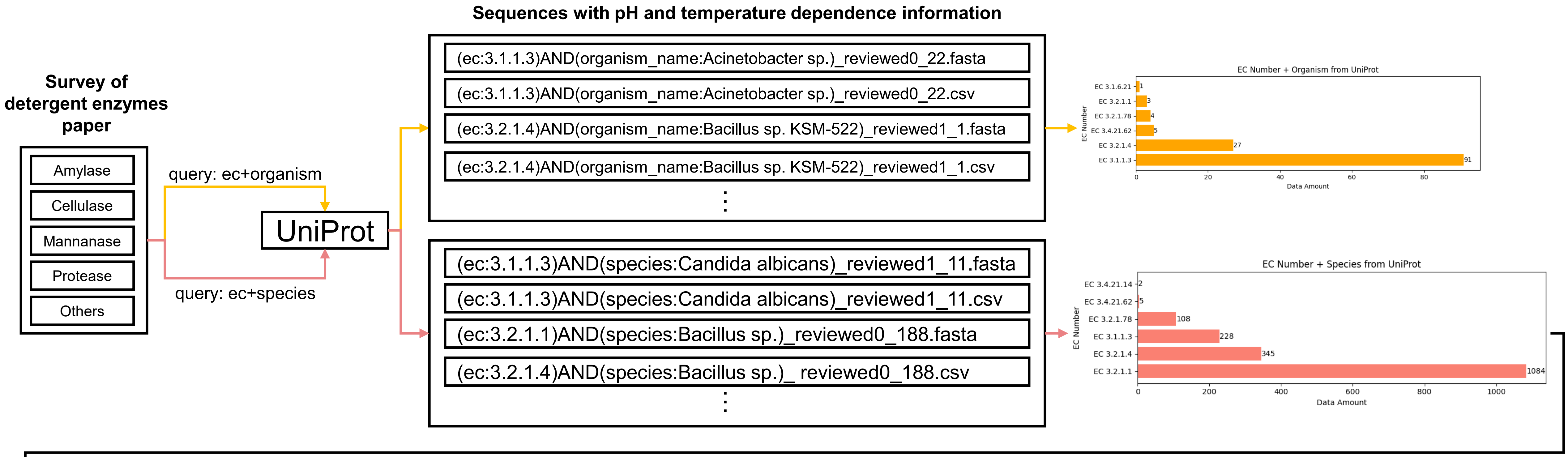
Clustering using CD-HIT

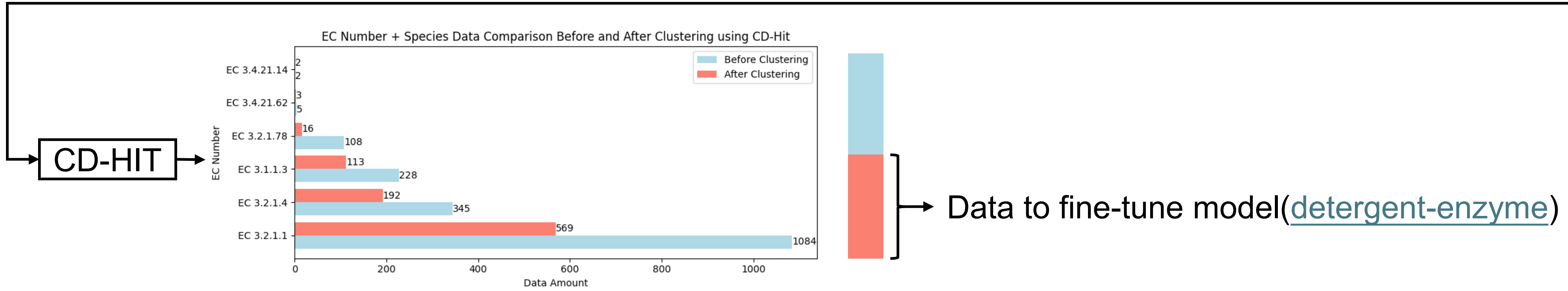
To identify representative sequences which will be positioned first in the MSA to enhance the generation

Data Collection



Data Collection





Data Pre-processing

```
>tr|A0A0A0N5B3|A0A0A0N5B3_STRRN alpha-amylase OS=Streptomyces rapamycinicus (strain ATCC 29253 / DSM 41530 / NRRL 5491 / AYB-994) OX=1343740 GN=D3C57_130380 PE=4 SV=1
MGGTTGTTGATVPKRRESHEHGGYEPAGPPAHATDDGTEAGGAGAGAAPRPADGFFTDHASWRWCFYVNVPFGLLTLAVVAAVLKVPEPPRRARFDYLGALFLSLFSTCAVLLTSWGGTEYAWGSRVILGLALGAAGSAILGHQPRAVSVTVGDRPVELRQDIELAGGAVLRGVVRATGG
RVVEDARVTLLDAAGNVVDTATTGPEGAFRFIDLSAGEYTVIAAGYPPVATVLQIAGGGRTERDLQLGYED
>tr|A0A0A0N5B3|A0ASG0N5B3_STRRN alpha-amylase OS=Streptomyces rapamycinicus (strain ATCC 29253 / DSM 41530 / NRRL 5491 / AYB-994) OX=1343740 GN=D3C57_130380 PE=4 SV=1
MGGTTGTTGATVPKRRESHEHGGYEPAGPPAHATDDGTEAGGAGAGAAPRPADGAAPAPAPLSPARVRMVFFGLMLALLLAALDQAVLTADGTPVREATVTLTDVRGEVVAATRSGREGGYVMDELVAGEYTLAASAPAFRPAALPVTVQAARETRQDIELAGGAVLRGVVRATGGRVV
EDARVTLLDAAGNVVDTATTGPEGAFRFIDLSAGEYTVIAAGYPPVATVLQIAGGGRTERDLQLGYED
>tr|LE5B3|LE5B3_STRRN alpha-amylase OS=Streptomyces rapamycinicus (strain ATCC 29253 / DSM 41530 / NRRL 5491 / AYB-994) OX=1343740 GN=D3C57_130380 PE=4 SV=1
MGGTTGTTGATVPKRRESHEHGGYEPAGPPAHATDDGTEAGGAGAGAAPRPADGAAPAPAAGVPVCGTVQHHDGTSVPRAALTLIDVGGRQIGRGATGEDGRYALSTPGAGSYVLIAAAGGHQPRARPAALPVTVQAARETRQDIELAGGAVLRGVVRATGGRVVEDARVTLLDAAG
NVVDTATTGPEGAFRFIDLSAGEYTVIAAGYPPVATVLQIAGGGRTERDLQLGYED
```

Multiple Sequence Alignment

Use Clustal Omega to do MSA on
- representative sequences with same EC number
- representative sequences with same species

These data will be input for conditional_generation_msa.py and train-msa.py."

Motif Position Scanning

Motif positions were identified using InterProScan with the Protein family (Pfam) database, which contains the most extensive entry collection.

detergent-paper

Datasets: lun610200/**detergent-papers** like 0 | Dataset card | **Viewer** | Files and versions | Community | Setting

Split (6)
amylase · 27 rows

Search this dataset

EC Number string · *classes* 1 value	Protein Name string · *classes* 1 value	Source Organism string · *lengths* 15 36	pH Optimum float64 4 45 ⌀	Temperature Optimum float64 20 200	Biotechnology Use string · *lengths* 9 89	Reference string · *lengths* 181 385
3.2.1.1	α-amylase	Pseudomonas stutzeri AS22	8	30	detergent industry	Maâlej, Hana, Noomen Hmidet, Olfa Ghorbel-Bellaaj and Monçef Nasri. “Purification an
3.2.1.1	α-amylase	Bacillus cohnii US147	9	45	starch saccharification, detergent industry	Ghorbel, Raoudha Ellouze, Sameh Maktouf, Ezedine Ben Massoud, Samir Bejar and Semia
3.2.1.1	α-amylase	Bacillus licheniformis AS08E	45	200	laundry industry	Roy, Jetendra Kumar and Ashis Kumar Mukherjee. “Applications of a high maltose
3.2.1.1	α-amylase	Bacillus sp. strain TSCVKK	8	30	detergent industry	Kiran, Kondepudi Kanthi and T. S. Chandra. “BIOTECHNOLOGICALLY RELEVANT ENZYMES AND…
3.2.1.1	α-amylase	Bacillus subtilis AS-S01a	6	55	commercial laundry detergents	Roy, Jetendra Kumar, Sudhir Kumar Rai and Ashis Kumar Mukherjee. “Characterization a
3.2.1.1	α-amylase	Bacillus subtilis DM-03	8	37	commercial laundry detergent	Mukherjee, Ashis Kumar, Munindra Borah and Sudhir Kumar Rai. “To study the influence
3.2.1.1	α-amylase	Nesterenkonia sp. strain F	7	33	detergent stable	Shafiei, Mohammad, Abed Ali Ziaee and Mohammad Ali Amoozegar. “Purification and…
3.2.1.1	α-amylase	Streptomyces strain A3	9	45	detergent and pharmaceutical industry	Chakraborty, Samrat, Ghanshyam Raut, Abhij Khopade, Kakasaheb Ramoo Mahadik and…
3.2.1.1	α-amylase	Bacillus licheniformis RA31	8	70	detergent industry	“Purification, characterization and potent detergent industry application of a…
null	null	Bacillus subtilis PF1	6	70	laundry detergents	Bhange, Khushboo, Venkatesh Chaturvedi and Renu Bhatt. “Simultaneous production of…
null	null	Aspergillus niger	4	50	detergent	Mitidieri, Sydnei, Anne Helene Souza

detergent-motif

Datasets: lun610200/detergent-motif like 0

Dataset card | Viewer | Files and versions | Community 1 | Settings

Split (6)
3.2.1.78 · 124 rows

Search this dataset

EC string · classes 1 value	Entry string · lengths 6 10	Protein string · classes 14 values	Species string · classes 18 values	Organism string · lengths 11 141	Motif_start float64 1 688 ∅	Motif_end float64 95 843 ∅	pH_left float64 5 6 ∅	pH_right float64 6 7 ∅	Temperature_left float64 50 50 ∅	Temperature_right float64 55 60 ∅	Sequence string · lengths 29 847
3.2.1.78	Q5PSP8	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	32	350	5	6	50	55	MFKKHTISLLILFLLASAVLAKPIEAHTVSPVNP
3.2.1.78	A0A0C5CFP8	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	30	348	5	6	50	55	MLKKLAVCLSIILLLLGAASPISAHTVYPVNPNA
3.2.1.78	A0A0H4TJH7	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	32	350	5	6	50	55	MFKKHTISLLIIFLLASAVLAKPIEAHTASPVNP
3.2.1.78	A0A8A6IMV9	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	37	355	5	6	50	55	MGELHLFKKHTISLLILFLLASAVLAKPIEAHTV
3.2.1.78	F1DT03	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	30	348	5	6	50	55	MLKKLAVCLSIVLLLLGAASPISAHTVYPVNPNA
3.2.1.78	Q2L7F9	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	37	355	5	6	50	55	MEELHLFKKHTISLLILFLLASAVLAKPIEAHTV
3.2.1.78	S5ZIG7	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	32	350	5	6	50	55	MFKKHTISLLIIFLLTSAVLAKPIEAHTVSPVNP
3.2.1.78	A0A5D4N083	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	32	350	5	6	50	55	MFKKHTICLLILFLLVSAALAKPIEAHTVPPVNP
3.2.1.78	D0UZJ9	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	32	350	5	6	50	55	MLKKHTISLLIIFLLASAVLAKPIEAHTVAPVNP
3.2.1.78	F2VQ62	Mannan endo-1,4-beta-mannosidase…	Bacillus subtilis	Bacillus subtilis	32	350	5	6	50	55	MFKKHTISLLIIFLLASAVLAQPIEAHTVSPVNP

detergent-enzyme

Datasets: lun610200 / detergent-enzyme | like 0 | Dataset card | Viewer | Files and versions | Community 1

Split (2)
train · 514 rows

Search this dataset

id	ec	protein	species	organism	motif_start	motif_end	pH_left	pH_right	temp_left	temp_right	sequence	pdb_data
string · lengths	string · classes	string · classes	string · lengths	string · lengths	float64	float64	float64	float64	float64	float64	string · lengths	string · lengths
6 – 10	6 values	89 values	12 – 32	11 – 141	1 – 988	26 – 1.14k	2.5 – 10	6 – 11	15 – 65	25 – 75	40 – 1.3k	18.4k – 811k
A0A1Q5KI96	3.2.1.1	alpha-amylase (EC 3.2.1.1)	Streptomyces sp.	Streptomyces sp. TSRI0107	42	435	null	null	null	null	tr|A0A1Q5KI96|A0A1Q5KI96_9ACTN MGGSTLVQDEPVAAAPPRLPASPAAPAAGLGPRRVRLVFFALMLALLLA…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AMY
A0A0M4FQR3	3.2.1.1	Alpha-amylase (EC 3.2.1.1)	Bacillus sp.	Bacillus sp. FJAT-22090	28	373	null	null	null	null	tr|A0A0M4FQR3|A0A0M4FQR3_9BACI MERKWWKEAVCYQVYPRSFMDSNGDGIGDLRGIISKLDYLKNLGVDVIW…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AMY
L8WJA6	3.2.1.1	alpha-amylase (EC 3.2.1.1)	Thanatephorus cucumeris	Thanatephorus cucumeris (strain…	362	439	null	null	null	null	tr|L8WJA6|L8WJA6_THACA MRTSLSFVVLLAVSANAARRSSHLHAAQRMHRRAPNSDKSGIPFQQPCK…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AMY
B0XUD0	3.2.1.1	alpha-amylase (EC 3.2.1.1)	Aspergillus fumigatus	Aspergillus fumigatus (strain…	65	389	6.5	null	37	null	tr|B0XUD0|B0XUD0_ASPFC MIYSLRILQWLCITAGLDLLVPSLAADTSAWKSRSIYQTMTDRFARTDG…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AMY
Q5AZF6	3.2.1.1	Alpha-amylase (EC 3.2.1.1)	Emericella nidulans	Emericella nidulans (strain FGSC A4 /…	60	361	null	null	null	null	tr|Q5AZF6|Q5AZF6_EMENI MRRLTCLSAFFLASASVAVASPSAEQWAERSIYQVMTDRFARPAGSPGD…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AMY
A0A2S0K0S6	3.2.1.4	Cell wall anchor domain-containing protein (EC…	Lysinibacillus sphaericus	Lysinibacillus sphaericus (Bacill…	792	867	null	null	null	null	tr|A0A2S0K0S6|A0A2S0K0S6_LYSSH MDIFKKQIHLVIIFILIFSLLSPAATGKAQHLEGETYEQNTSDINNTEL…	HEADER 01-JUN-22 TITLE A PREDICTION FOR UNCHARACT
Q59445	3.2.1.4	Endoglucanase 3 (EC 3.2.1.4)	Fibrobacter succinogenes	Fibrobacter succinogenes…	338	640	5.3	null	25	null	tr|Q59445|Q59445_FIBSU MNFKTKSSLALSAAALLFLSLCGDDSSKSAAPDEPAVNPTDSTTIPTPL…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ENDOGLUCA
A0A2A8V8N3	3.2.1.1	alpha-amylase (EC 3.2.1.1)	Bacillus sp.	Bacillus sp. AFS015802	66	360	null	null	null	null	tr|A0A2A8V8N3|A0A2A8V8N3_9BACI MKRRVLSLILVPFLLFYALPVGAVEKEERKWQDETIYFLMVDRFNNGDM…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AML
Q65MP4	3.2.1.78	Mannan endo-1,4-beta-mannosidase (EC 3.2.1.78)	Bacillus licheniformis	Bacillus licheniformis…	30	348	6	7	50	60	tr|Q65MP4|Q65MP4_BACLD MKKNIVCSIFALLLAFAVSQPSYAHTVSPVNPNAQPTTKAVMNWLAHLP…	HEADER 01-JUN-22 TITLE A PREDICTION FOR MANNAN EN
A7UG67	3.2.1.4	Endoglucanase (EC 3.2.1.4)	Fibrobacter succinogenes	Fibrobacter succinogenes (stra…	115	551	5.3	null	25	null	tr|A7UG67|A7UG67_FIBSS MKLNFGLKQYIPLAATVLSLATVANAATAYINQIGYRPADPKEFALVDG…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ENDOGLUCA
A2QTZ0	3.1.1.3	Contig An09c0100, genomic contig (EC 3.1.1.3)	Aspergillus niger	Aspergillus niger (strain ATCC MYA-…	59	546	2.5	null	25	35	tr|A2QTZ0|A2QTZ0_ASPNC MATPRMGGVVAAVTVAIVLFAIYLSPSTDVIVEHINDYYNVVFPKAPVS…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ASPERGILL
A0A6C0Q9Z6	3.2.1.78	Mannan endo-1,4-beta-mannosidase (EC 3.2.1.78)	Streptomyces sp.	Streptomyces sp. S4.7	48	340	null	null	null	null	tr|A0A6C0Q9Z6|A0A6C0Q9Z6_9ACTN MRGNRTRPRPRSRPRFRLVALATAALTTAVVSLTASPAPAGAFPATPKQ…	HEADER 01-JUN-22 TITLE A PREDICTION FOR MANNAN EN
A1C4I6	3.2.1.1	Maltase MalT	Aspergillus clavatus	Aspergillus clavatus (strain ATCC 1007 …	39	400	null	null	null	null	tr|A1C4I6|A1C4I6_ASPCL MSPHALREVPTGTNNWWKEATVYQVYPASFKDSNGDGWGDIPGLISKIP…	HEADER 01-JUN-22 TITLE A PREDICTION FOR MALTASE M
A0A7X8N2B3	3.2.1.1	Alpha-amylase (EC 3.2.1.1)	Bacillus sp.	Bacillus sp. R01	65	405	null	null	null	null	tr|A0A7X8N2B3|A0A7X8N2B3_9BACI MKFENRIVSFVLTILLAVAVLIPYTSEPAKAHHNGTNGTLMQYFEWYLP…	HEADER 01-JUN-22 TITLE A PREDICTION FOR ALPHA-AMY

Learning method

- Order Agnosic Mask Collater for masking batch data according to Hoogeboom et al. OA ARDMS

timestep	8																		
Mask	True	True	False	True	True	False	True	False	True	False	False	False	True	False	True	False	False	False	False
Src	28	28	15	28	28	4	28	7	28	1	5	17	28	4	28	0	30	30	30
~~Tokenized~~ Tgt	17	13	15	8	4	4	5	7	0	1	5	17	0	4	4	0	30	30	30
Input_mask	True	True	True	True	True	True	True	True	True	True	True	True	True	True	True	True	False	False	False

Learning method

- Order Agnostic Mask Collater for masking batch data according to Hoogeboom et al. OA ARDMS

loss tensor([0.1043, 2.1216, 3.0950, 2.8898, 3.0310, 3.0450, 2.7027, 2.5332], device='cuda:1', grad_fn=<SliceBackward0>)

timestep $\boxed{8}$

Reweight term 1. / 8 = 0.125

Reweight term	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125

nonpad_tokens = 16

True	True	True	True	True	True	True	True	True	True	True	True	True	True	True	True	False	False	False

_n_tokens	16	16	16	16	16	16	16	16

Ce_loss (cross entropy loss) = _n_tokens * reweight term * loss

Ce_loss	0.2086	4.2432	6.19	5.7796	6.062	6.09	5.4054	5.0664

Learning method

- Reward weight

Output of the model to softmax

Convert the model output into a probability distribution

Loss based on hydrogen and disulfide bridges

Calculate probability of target amino acid in output of softmax and calculate negative log loss

Calculate the reward loss

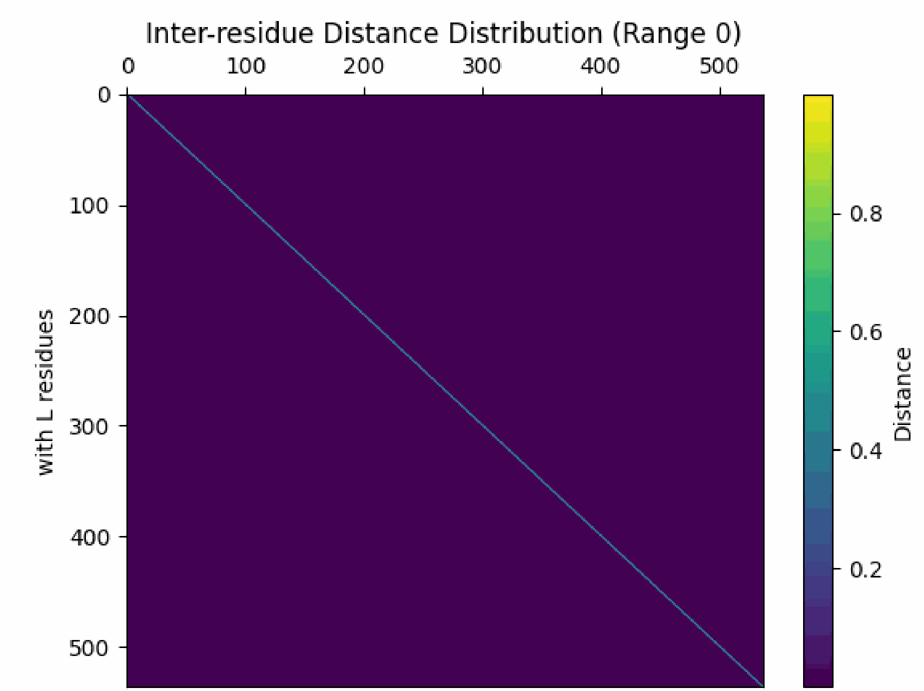
Add the reward loss on top of the original Order Agnostic Mask

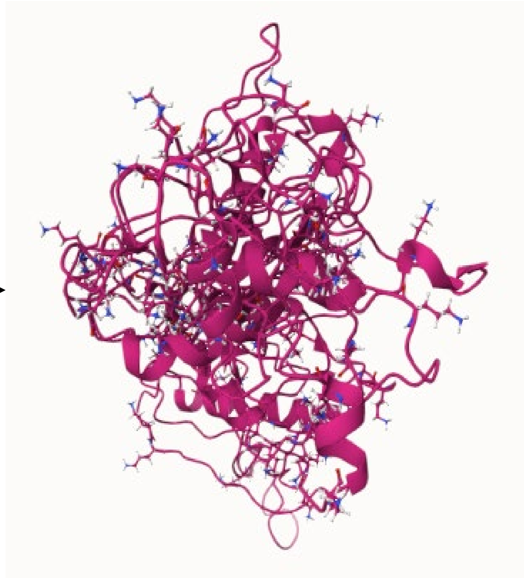
2 Experiments

Predict structure

- ProFOLD

MDBEHKEEBGFE………ZBFTLQFNGSFKK
MMNGQWVLPRIF………VYQQGKLVWGIEP
MMNGQWVLPRIA……....EYGANKLVWGIEP
MKKKGLKKTFFVI………---------------------------

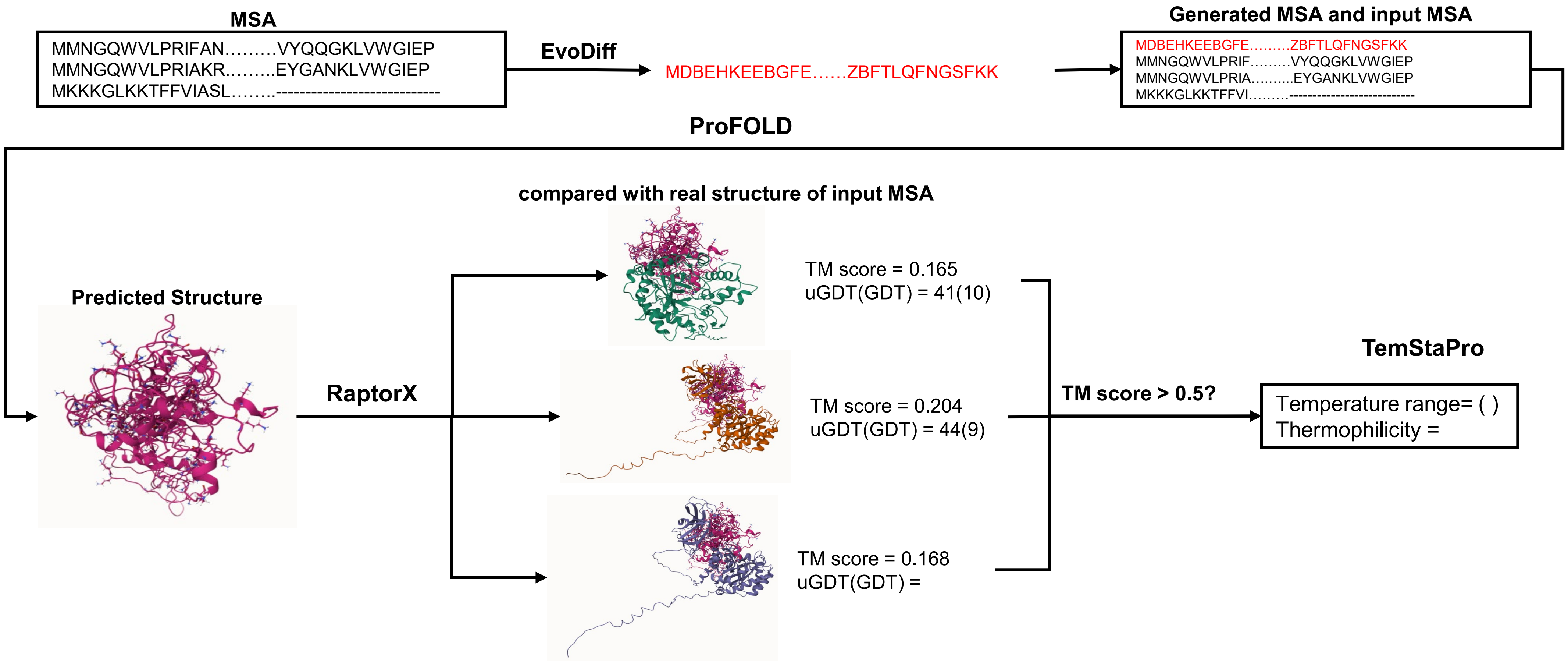




EC 3.2.1.4 CMCase from Paenibacillus sp. IHB B 3084

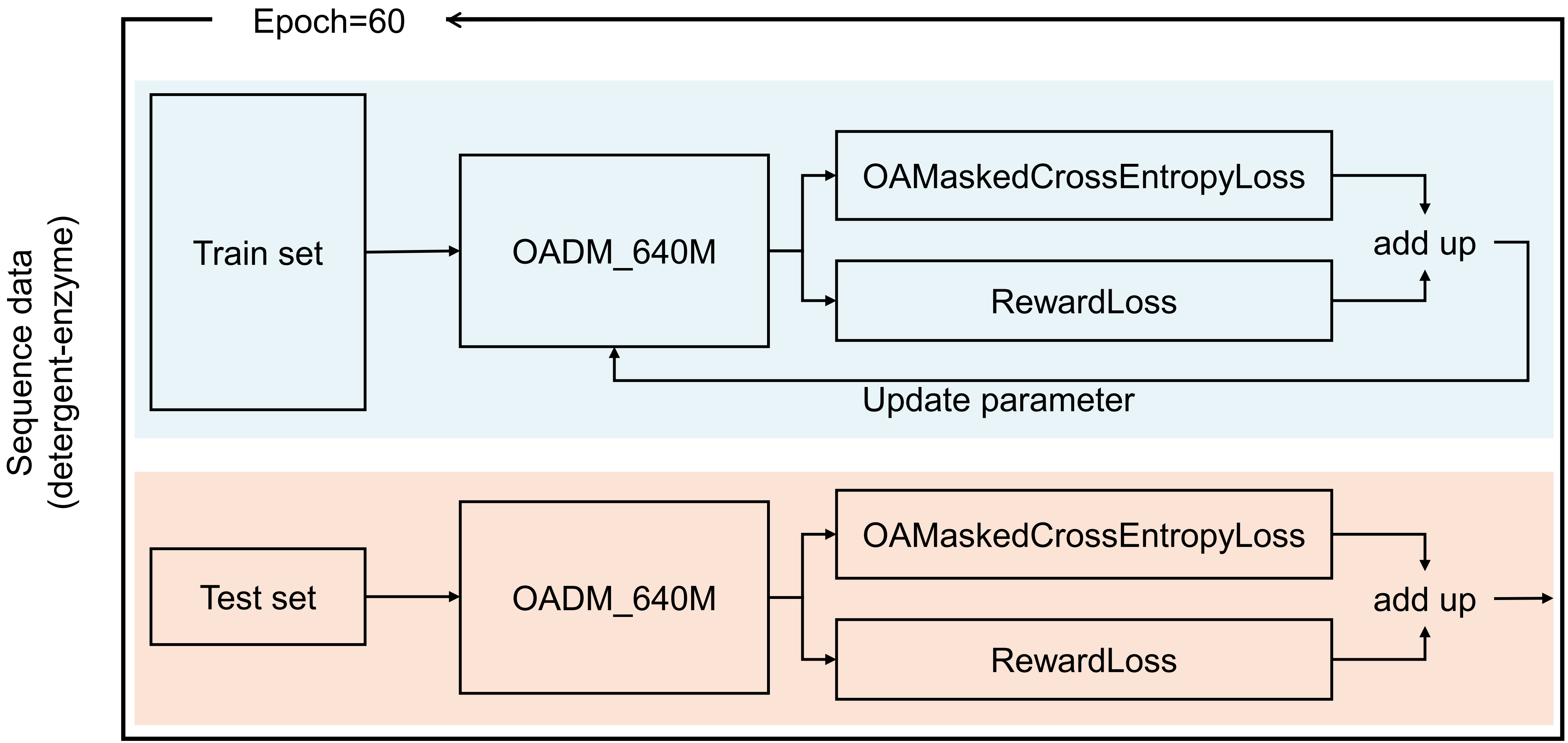
Predict structure

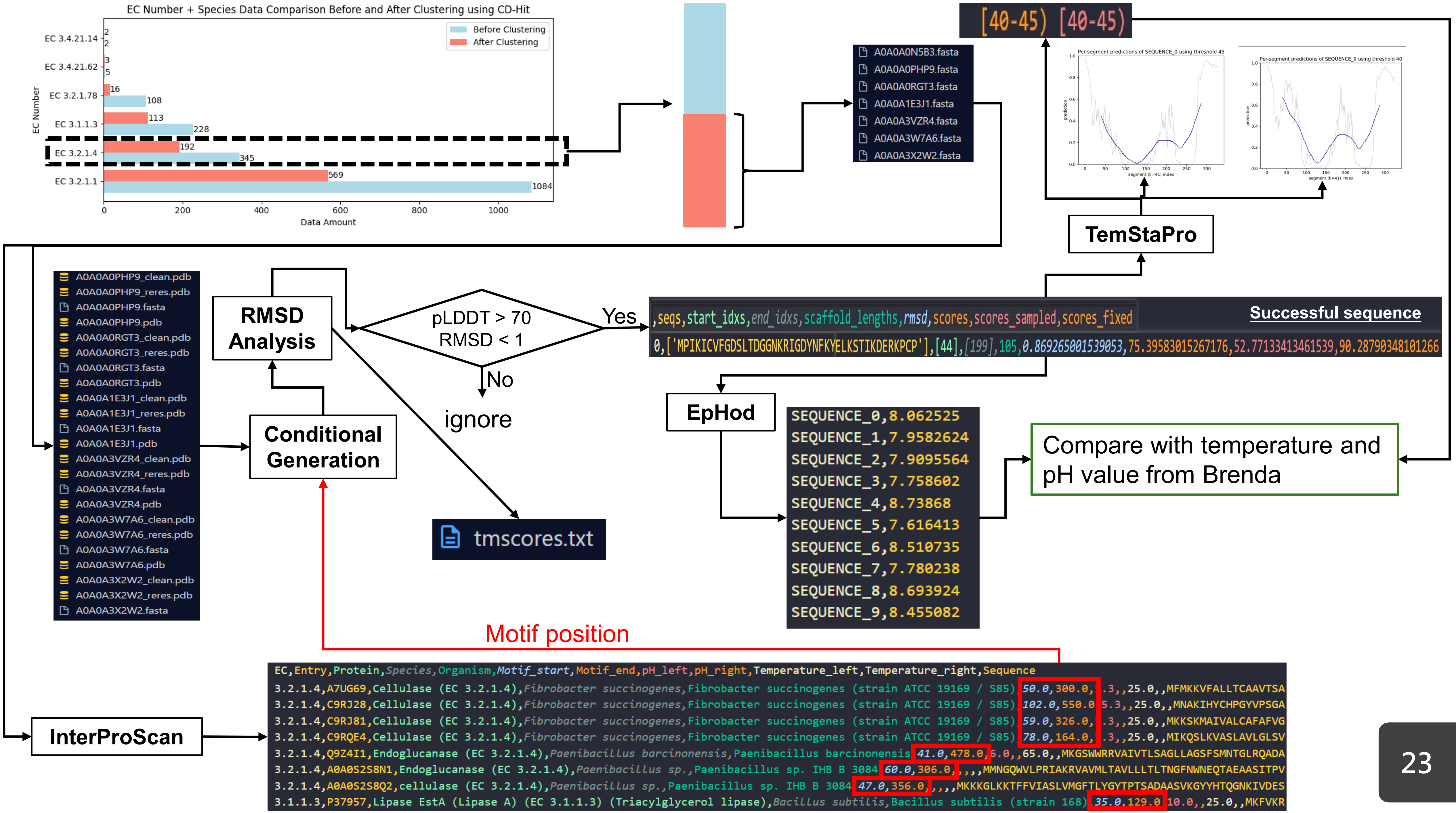
- ProFOLD



EC 3.2.1.4 CMCase from Paenibacillus sp. IHB B 3084

Fine-tuning EvoDiff





EC Number + Species Data Comparison Before and After Clustering using CD-Hit
Before Clustering
After Clustering
EC Number
EC 3.4.21.14
EC 3.4.21.62
EC 3.2.1.78
EC 3.1.1.3
EC 3.2.1.4
EC 3.2.1.1
2
2
3
5
16
108
113
228
192
345
569
1084
0
200
400
600
800
1000
Data Amount
A0A0A0N5B3.fasta
A0A0A0PHP9.fasta
A0A0A0RGT3.fasta
A0A0A1E3J1.fasta
A0A0A3VZR4.fasta
A0A0A3W7A6.fasta
A0A0A3X2W2.fasta
[40-45) [40-45)
Per-segment predictions of SEQUENCE_0 using threshold 45
Per-segment predictions of SEQUENCE_0 using threshold 40
TemStaPro
A0A0A0PHP9_clean.pdb
A0A0A0PHP9_reres.pdb
A0A0A0PHP9.fasta
A0A0A0PHP9.pdb
A0A0A0RGT3_clean.pdb
A0A0A0RGT3_reres.pdb
A0A0A0RGT3.fasta
A0A0A0RGT3.pdb
A0A0A1E3J1_clean.pdb
A0A0A1E3J1_reres.pdb
A0A0A1E3J1.fasta
A0A0A1E3J1.pdb
A0A0A3VZR4_clean.pdb
A0A0A3VZR4_reres.pdb
A0A0A3VZR4.fasta
A0A0A3VZR4.pdb
A0A0A3W7A6_clean.pdb
A0A0A3W7A6_reres.pdb
A0A0A3W7A6.fasta
A0A0A3W7A6.pdb
A0A0A3X2W2_clean.pdb
A0A0A3X2W2_reres.pdb
A0A0A3X2W2.fasta
RMSD Analysis
pLDDT > 70
RMSD < 1
Yes
No
ignore
Successful sequence
,seqs,start_idxs,end_idxs,scaffold_lengths,rmsd,scores,scores_sampled,scores_fixed
0,['MPIKICVFGDSLTDGGNKRIGDYNFKYELKSTIKDERKPCP'],[44],[199],105,0.869265001539053,75.39583015267176,52.77133413461539,90.28790348101266
Conditional Generation
tmscores.txt
EpHod
SEQUENCE_0,8.062525
SEQUENCE_1,7.9582624
SEQUENCE_2,7.9095564
SEQUENCE_3,7.758602
SEQUENCE_4,8.73868
SEQUENCE_5,7.616413
SEQUENCE_6,8.510735
SEQUENCE_7,7.780238
SEQUENCE_8,8.693924
SEQUENCE_9,8.455082
Compare with temperature and pH value from Brenda
Motif position
InterProScan
EC,Entry,Protein,Species,Organism,Motif_start,Motif_end,pH_left,pH_right,Temperature_left,Temperature_right,Sequence
3.2.1.4,A7UG69,Cellulase (EC 3.2.1.4),Fibrobacter succinogenes,Fibrobacter succinogenes (strain ATCC 19169 / S85),50.0,300.0,5.3,,25.0,,MFMKKVFALLTCAAVTSA
3.2.1.4,C9RJ28,Cellulase (EC 3.2.1.4),Fibrobacter succinogenes,Fibrobacter succinogenes (strain ATCC 19169 / S85),102.0,550.0,5.3,,25.0,,MNAKIHYCHPGYVPSGA
3.2.1.4,C9RJ81,Cellulase (EC 3.2.1.4),Fibrobacter succinogenes,Fibrobacter succinogenes (strain ATCC 19169 / S85),59.0,326.0,5.3,,25.0,,MKKSKMAIVALCAFAFVG
3.2.1.4,C9RQE4,Cellulase (EC 3.2.1.4),Fibrobacter succinogenes,Fibrobacter succinogenes (strain ATCC 19169 / S85),78.0,164.0,5.3,,25.0,,MIKQSLKVASLAVLGLSV
3.2.1.4,Q9Z4I1,Endoglucanase (EC 3.2.1.4),Paenibacillus barcinonensis,Paenibacillus barcinonensis,41.0,478.0,6.0,,65.0,,MKGSWWRRVAIVTLSAGLLAGSFSMNTGLRQADA
3.2.1.4,A0A0S2S8N1,Endoglucanase (EC 3.2.1.4),Paenibacillus sp.,Paenibacillus sp. IHB B 3084,60.0,306.0,,,,,MMNGQWVLPRIAKRVAVMLTAVLLLTLTNGFNWNEQTAEAASITPV
3.2.1.4,A0A0S2S8Q2,cellulase (EC 3.2.1.4),Paenibacillus sp.,Paenibacillus sp. IHB B 3084,47.0,356.0,,,,,MKKKGLKKTFFVIASLVMGFTLYGYTPTSADAASVKGYYHTQGNKIVDES
3.1.1.3,P37957,Lipase EstA (Lipase A) (EC 3.1.1.3) (Triacylglycerol lipase),Bacillus subtilis,Bacillus subtilis (strain 168),35.0,129.0,10.0,,25.0,,MKFVKR

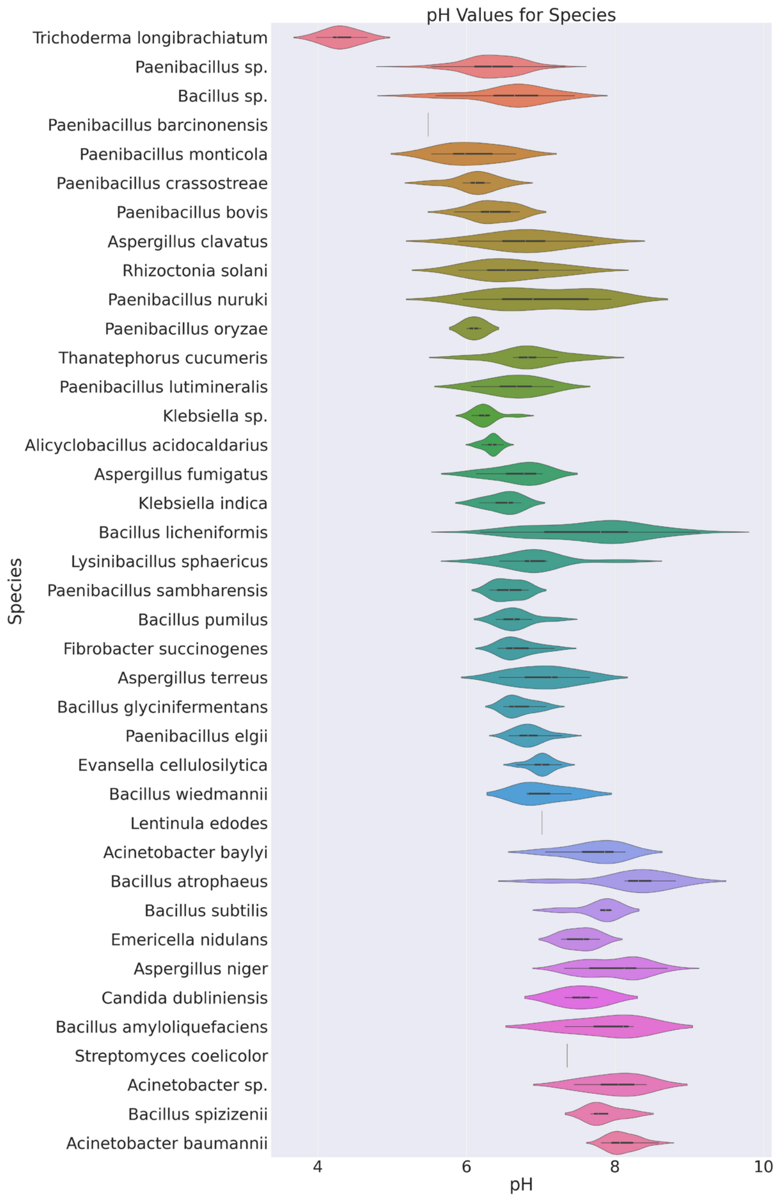
3 Results

Evaluation of successful sequence

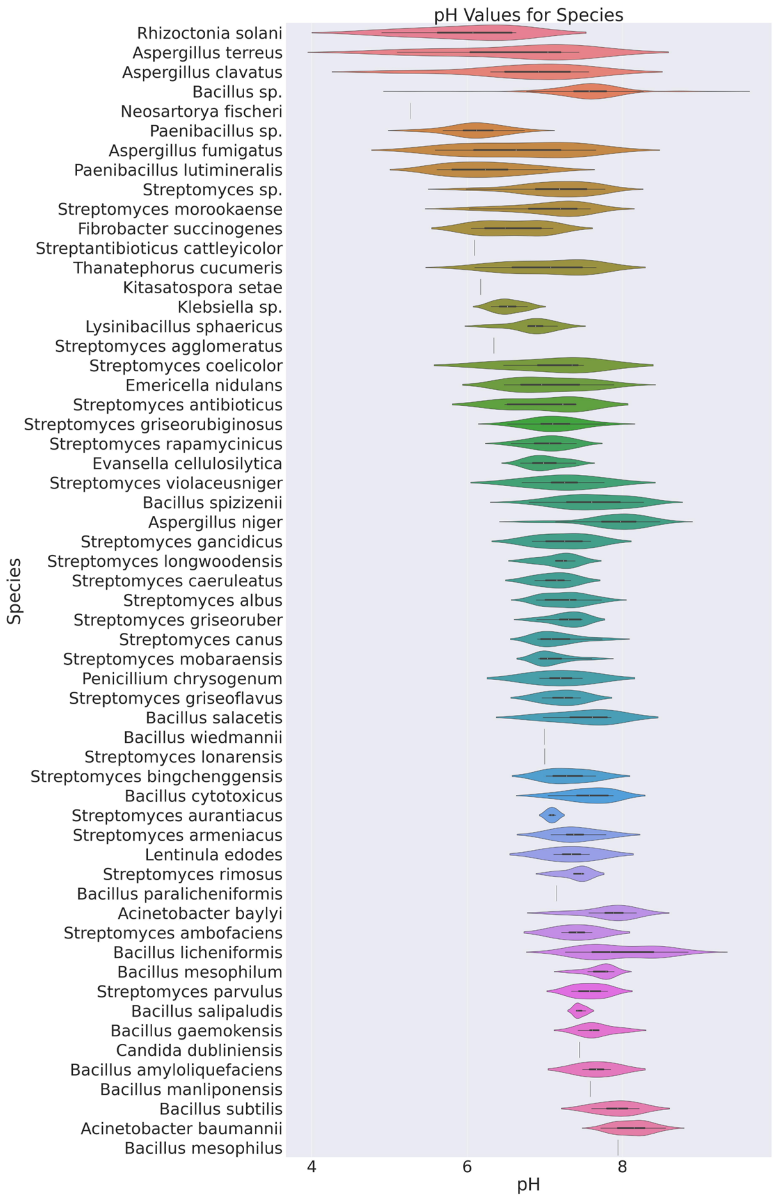
metrics	reference	generated	How	criteria	code
Motif RMSD	{pdb}_reres.pdb	SEQUENCE_{i}.pdb	Use RMSD to compare the ranges of the original motif and the new motif match	(motif_rmsd < 1) and (confidence score > 70)	rmsd_analyasis.py
Motif confidence score		SEQUENCE_{i}.pdb	Calculate the mean B-factor(here is pLDDT) of every atom from using PDBParser in residues within the range of the motif match.	same as above	rmsd_analyasis.py
Similarity	Sequence used to generate	Motif in generated sequence	`difflib.SequenceMatcher(None,seq1, seq2)` to compare sequences	[0, 1]	rmsd_analyasis.py
TM score	{pdb}_reference.pdb	SEQUENCE_{i}.pdb	Compare structure	>0.5 in about same fold	conditional_generation_msa.py

Prediction of pH value

Pretrained model

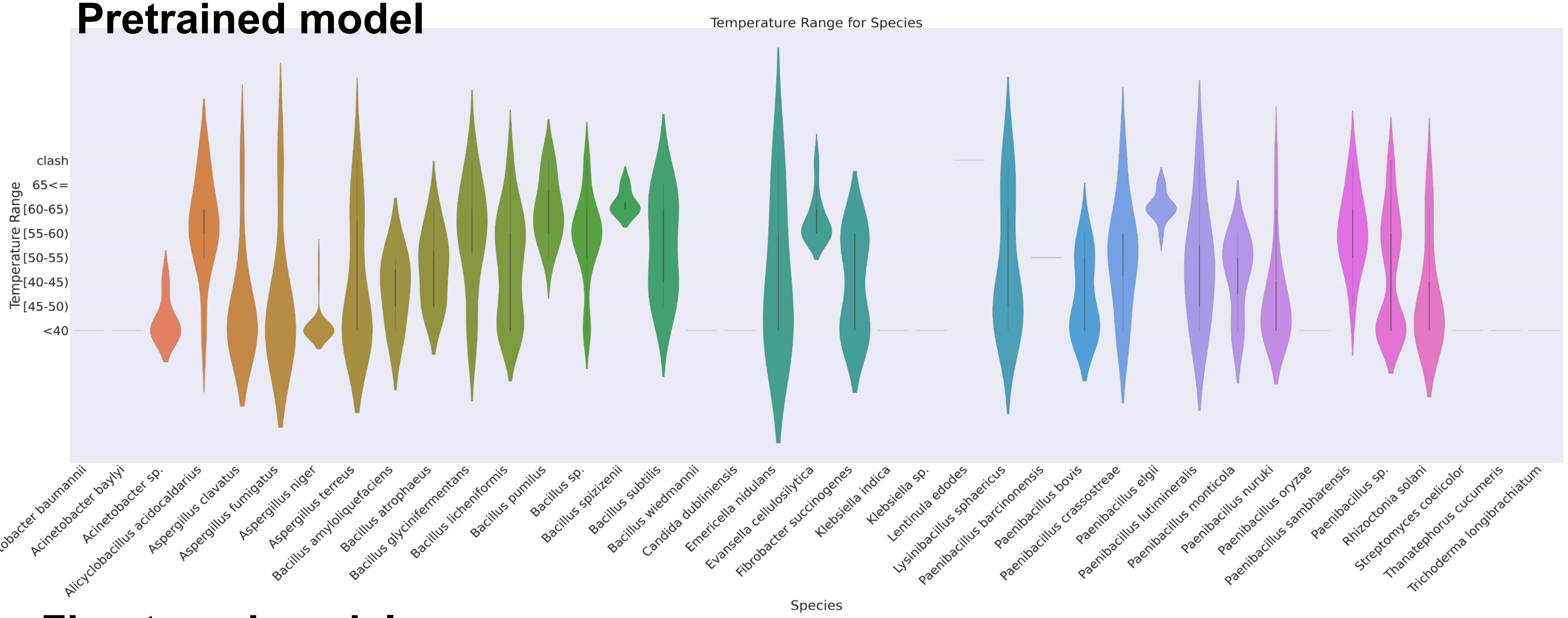


Fine-tuned model

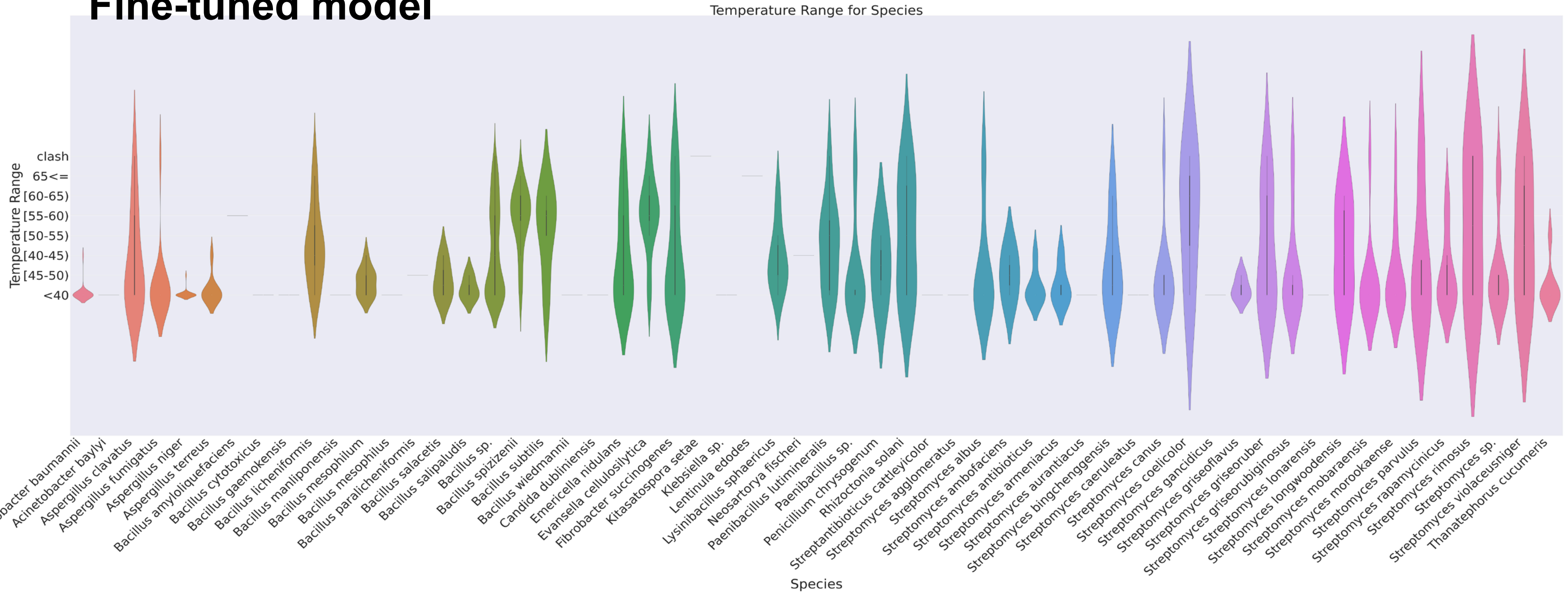


Prediction of temperature range

Pretrained model



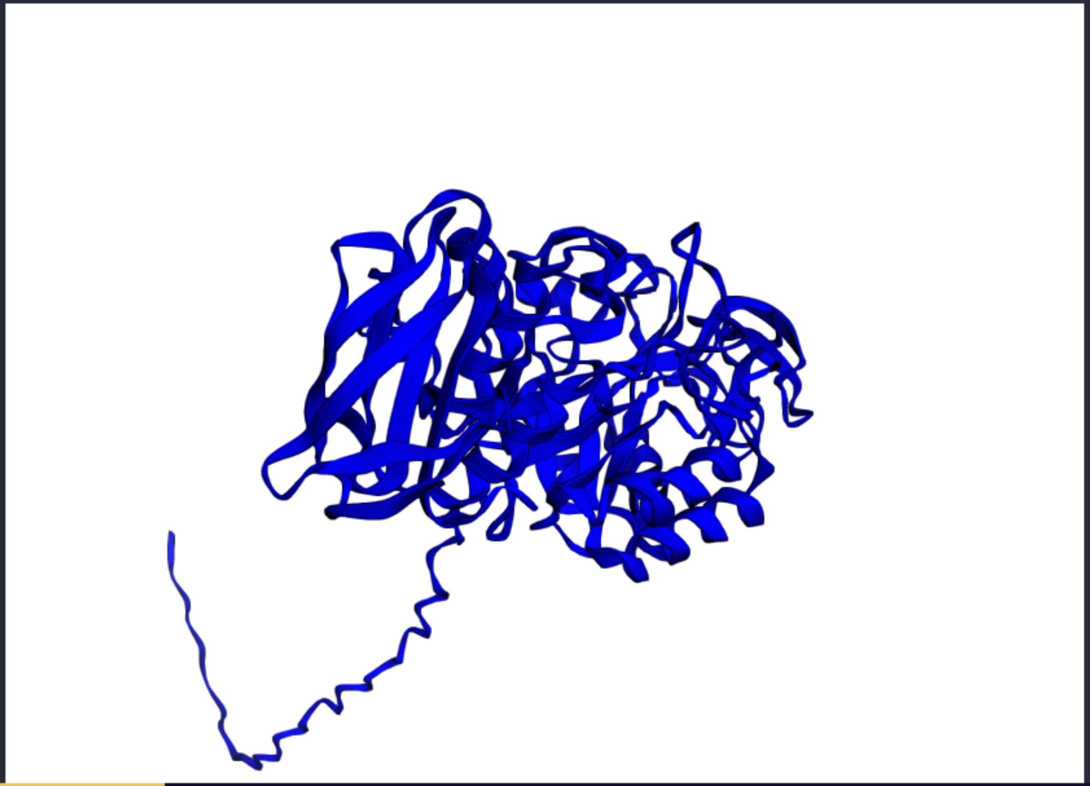
Fine-tuned model



Visualization

```python
import py3Dmol
import os

dir = 'data/scaffolding-pdbs'
pdb = 'L8ETE9'

view = py3Dmol.view(js='https://3dmol.org/build/3Dmol.js')
view.addModel(open(os.path.join(dir, f'{pdb}.pdb'),'r').read(),'pdb')

view.setStyle({'cartoon': {'colorscheme': {'prop':'b','gradient': 'roygb','min':0.5,'max':0.9}}}) # as color is set to lDDT
# view.setStyle({'cartoon': {'color':'spectrum'}})

view.zoomTo()
view.show()
```

✓ 0.0s

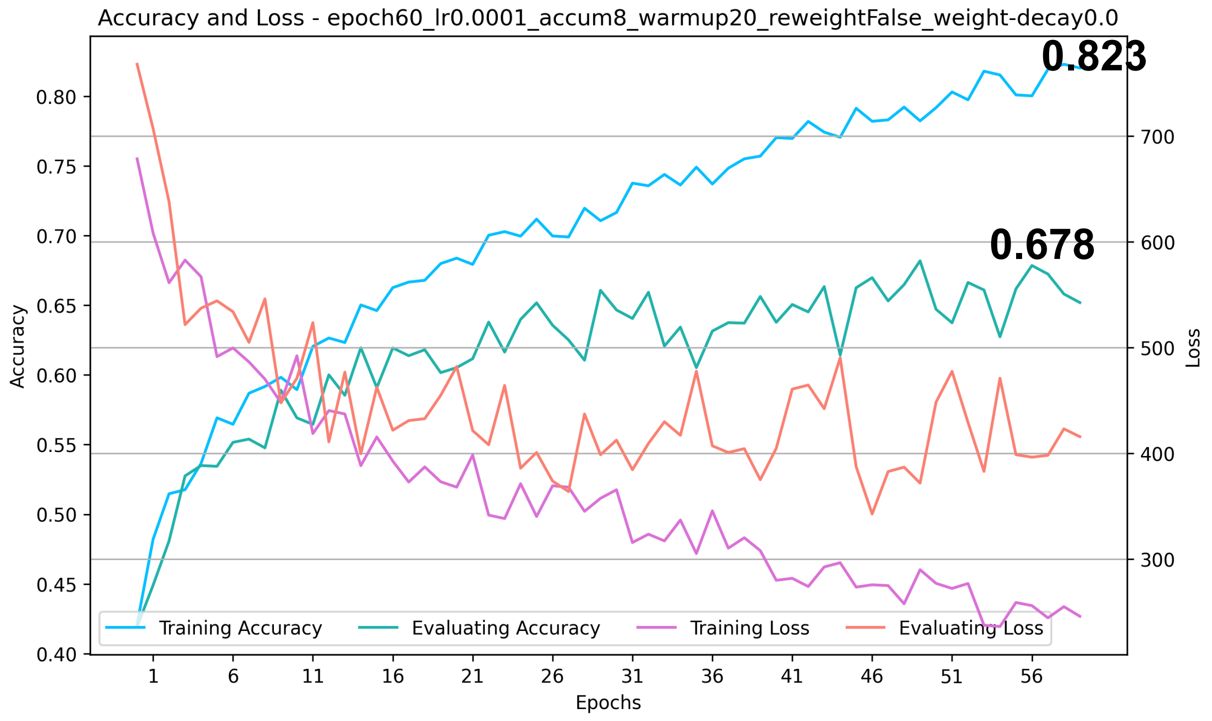


Fine-tuning EvoDiff

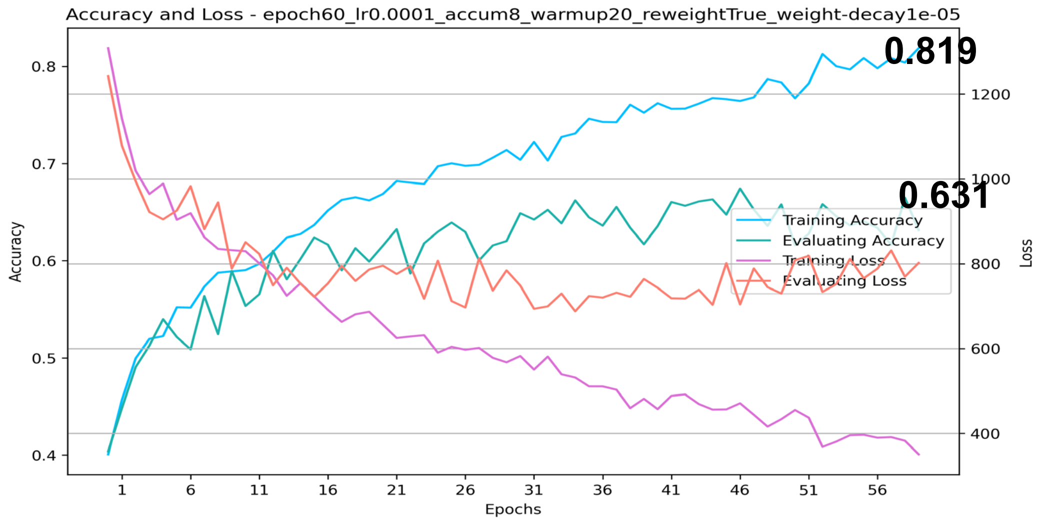
- Best configuration

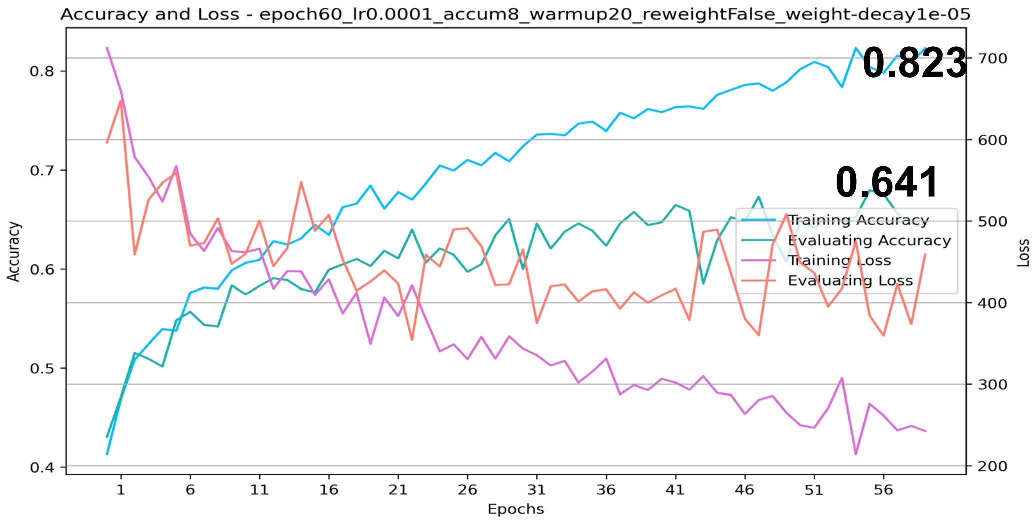
```
"gpus": 2,
"range": "species",
"random_seed": 42,
"lr": 0.0001,
"epochs": 60,
"train_batch_size": 1,
"validation_batch_size": 1,
"test_batch_size": 1,
"gradient_accumulation_steps": 8,
"warmup_steps": 20,
"weight_decay": 0,
"save_steps": 10,
"reweight": false,
"best_step": 56
```

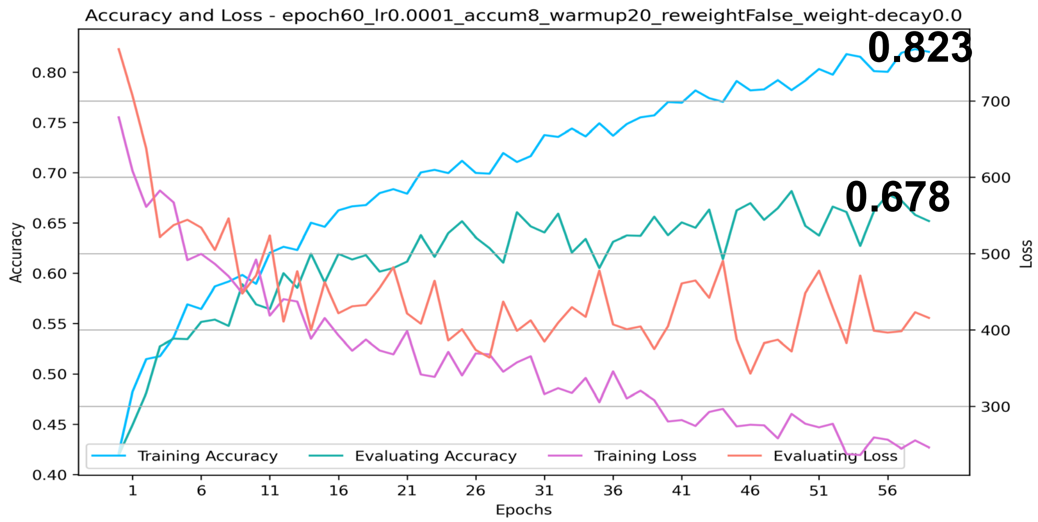
- Best accuracy



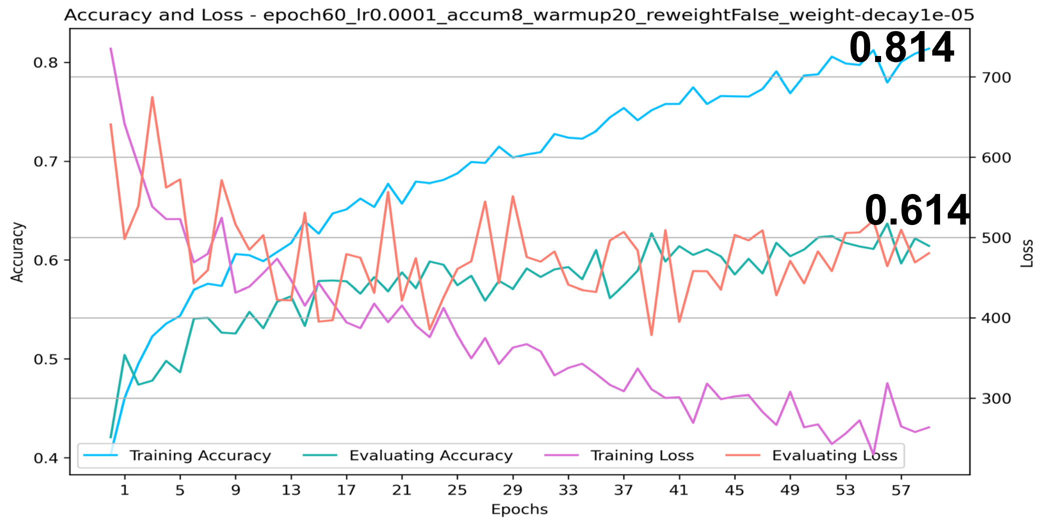
Fine-tuning EvoDiff







Reward mechanism



4 Conclusion

Repository

GitHub

Hugging Face

Summary

- **Creation of a Detergent-Enzyme Sequence Dataset**:

  Successfully created a dataset of detergent-compatible enzyme sequences, now available on Hugging Face Hub for public access and use in training models.

- **Fine-Tuning EvoDiff:**

  Successfully fine-tuned EvoDiff using custom datasets, guiding the model to generate sequences with stronger hydrogen bonds and disulfide bridges.

- **Expanded Knowledge and Application in Enzyme Engineering:**

  Gained valuable insights into various protein databases and related knowledge, while also learning how to apply machine learning techniques in enzyme engineering.

Future work

1. Convert predicted sequences to structures before calculating loss, focusing on alpha helices and beta sheets.
2. Validate whether the generated sequences exist in the real world and if they truly exhibit extreme heat resistance and detergent compatibility.
3. Assess whether the reward mechanism for disulfide bond formation effectively influences EvoDiff to generate sequences with more disulfide bridges.